

Comparative Analysis of a Massive MIMO System using MATLAB (MUSIC and ESPRIT)

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Abstract— This paper presents a comparative performance analysis of the MUSIC and ESPRIT algorithms for Direction of Arrival (DoA) estimation in massive Multiple-Input Multiple-Output (MIMO) systems under limited snapshot availability and spatially correlated source conditions. A MATLAB-based simulation framework is developed using a uniform linear array with varying antenna configurations to evaluate estimation accuracy, robustness, and computational efficiency. The impact of snapshot count, antenna array size, and spatial correlation on DoA estimation performance is analyzed using Root Mean Square Error (RMSE) and execution time metrics. Simulation results demonstrate that MUSIC achieves superior angular resolution and lower RMSE when sufficient snapshots are available, whereas ESPRIT exhibits greater robustness and reduced computational complexity under snapshot-constrained and correlated environments. The effectiveness of DoA estimation is further validated through adaptive beamforming using Least Mean Squares (LMS) and Sample Matrix Inversion (SMI) algorithms, confirming accurate beam steering and interference suppression in massive MIMO systems.

I. Introduction:

Massive Multiple-Input Multiple-Output (MIMO) technology has emerged as a key enabler for fifth- and sixth-generation (5G/6G) wireless communication systems due to its ability to significantly enhance spectral efficiency, system capacity, and interference suppression. By employing a large number of antenna elements at the base station, massive MIMO systems exploit spatial degrees of freedom to serve multiple users simultaneously within the same time–frequency resources.

Accurate Direction of Arrival (DoA) estimation plays a critical role in massive MIMO beamforming and spatial

multiplexing. High-resolution DoA estimation enables precise beam steering toward desired users while effectively suppressing interference and noise. Conventional beamforming techniques such as Bartlett and Capon offer low computational complexity but suffer from limited angular resolution, particularly in scenarios involving closely spaced sources or low signal-to-noise ratio (SNR) conditions.

Subspace-based algorithms, notably Multiple Signal Classification (MUSIC) and Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT), provide superior angular resolution and are widely adopted for DoA estimation. MUSIC achieves high accuracy by exploiting the orthogonality between signal and noise subspaces, whereas ESPRIT directly estimates signal directions using the rotational invariance property of array structures, avoiding spectral search. Despite their effectiveness, the performance of these algorithms in massive MIMO systems is strongly influenced by practical constraints such as limited snapshot availability, spatial correlation among antenna elements, and increased array dimensionality.

While numerous studies have investigated MUSIC and ESPRIT for conventional MIMO and small-scale antenna arrays, a systematic quantitative comparison of their performance in massive MIMO environments under limited snapshots and spatially correlated source conditions remains insufficiently explored. These conditions are highly relevant in practical deployments, where fast channel variations and correlated propagation environments restrict the number of available snapshots and degrade estimation accuracy.

Motivated by this gap, this paper presents a comprehensive MATLAB-based analysis of MUSIC and ESPRIT DoA estimation techniques in a massive MIMO framework. The study evaluates their performance under varying snapshot

counts, antenna array sizes, and spatial correlation levels. Furthermore, the impact of DoA estimation accuracy on adaptive beamforming using Least Mean Squares (LMS) and Sample Matrix Inversion (SMI) algorithms is examined to validate practical beam steering performance.

Contributions of This Paper:

The main contributions of this work are summarized as follows:

- Development of a MATLAB-based massive MIMO system model employing a uniform linear array for DoA estimation and beamforming analysis.
- Quantitative comparison of MUSIC and ESPRIT algorithms under limited snapshot conditions in massive MIMO scenarios.
- Investigation of spatial correlation effects on DoA estimation accuracy and resolution.
- Performance evaluation with varying antenna array sizes (32, 64, and 128 elements).

Validation of adaptive beamforming using LMS and SMI algorithms based on estimated DoAs.

II. Related Work and Literature Review

Direction of Arrival (DoA) estimation has been extensively studied as a fundamental signal processing problem in array-based wireless communication systems. With the emergence of massive MIMO as a key technology for 5G and beyond, accurate DoA estimation has gained renewed importance due to its direct impact on beamforming, spatial multiplexing, and interference mitigation.

A. Subspace-Based DoA Estimation Techniques:

Conventional DoA estimation methods such as Bartlett and Capon beamformers provide low-complexity solutions but suffer from limited angular resolution, especially in scenarios involving closely spaced sources or low signal-to-noise ratios. To overcome these limitations, subspace-based techniques have been widely adopted. Among them, MUSIC exploits the orthogonality between the signal and noise subspaces derived from the covariance matrix of the received signal, enabling high-resolution DoA estimation. ESPRIT, on the other hand, leverages the rotational invariance property of structured antenna arrays to estimate DoAs directly without spectral search, offering reduced computational complexity.

Numerous studies have demonstrated that MUSIC provides superior angular resolution compared to ESPRIT under ideal conditions. However, this improved resolution comes at the cost of increased computational burden and sensitivity to covariance matrix estimation errors, particularly in low snapshot scenarios.

B. DoA Estimation in Large-Scale and Massive MIMO Systems:

The extension of classical DoA estimation techniques to large-scale and massive MIMO systems introduces new challenges. The increased number of antenna elements enhances spatial resolution but also increases computational complexity and sensitivity to practical impairments. Several works have investigated the scalability of MUSIC and ESPRIT for large antenna arrays, highlighting that ESPRIT is more computationally efficient for massive arrays due to the absence of spectral scanning, while MUSIC benefits more from array size in terms of resolution.

Recent studies have also explored reduced-dimension and hybrid architectures to mitigate the complexity associated with massive MIMO DoA estimation. While these approaches are effective, they often assume ideal conditions or focus on specific frequency bands such as millimeter-wave or terahertz systems, limiting their applicability to general MATLAB-based system-level evaluations.

C. Impact of Limited Snapshots and Spatial Correlation:

In practical wireless environments, the number of available snapshots is often limited by channel time variations, user mobility, and latency constraints. Limited snapshot availability degrades covariance matrix estimation accuracy, adversely affecting subspace separation and DoA estimation performance. Prior research has shown that MUSIC is particularly sensitive to snapshot limitations, whereas ESPRIT demonstrates relatively stable performance under constrained data conditions.

Spatial correlation among antenna elements is another critical factor in massive MIMO systems, arising due to dense antenna placement, mutual coupling, and correlated propagation environments. Spatial correlation reduces the effective degrees of freedom of the array and degrades DoA estimation accuracy. While several studies acknowledge the adverse effects of correlation, a systematic comparison of MUSIC and ESPRIT under correlated massive MIMO scenarios remains limited.

D. Research Gap and Motivation:

Although MUSIC and ESPRIT have been widely analysed in conventional MIMO systems, there is a lack of comprehensive quantitative studies that jointly examine the impact of limited snapshots, spatial correlation, and antenna array scaling in massive MIMO environments. Most existing works either focus on idealized conditions or address only one of these practical constraints in isolation.

Motivated by this gap, the present work provides a detailed comparative analysis of MUSIC and ESPRIT DoA estimation algorithms in massive MIMO systems under realistic operating conditions. By evaluating performance across varying snapshot counts, antenna array sizes, and correlation levels, this study aims to offer practical insights into algorithm selection for future large-scale wireless communication systems.

III. System Model and Mathematical Formulation:

A. Massive MIMO System Model:

Consider a massive MIMO receiver equipped with a uniform linear array (ULA) consisting of M antenna elements, where $M \in \{32, 64, 128\}$. The antenna elements are equally spaced with an inter-element distance $d = \lambda/2$, where λ denotes the wavelength of the incoming signal.

Assume K narrowband far-field sources impinge on the array from directions $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_K]$. The baseband received signal vector at time instant n can be expressed as

$$\mathbf{x}(n) = \mathbf{A}(\boldsymbol{\theta})\mathbf{s}(n) + \mathbf{v}(n)$$

where $\mathbf{s}(n) \in \mathbb{C}^{K \times 1}$ is the source signal vector, $\mathbf{v}(n)$ represents additive white Gaussian noise (AWGN), and $\mathbf{A}(\boldsymbol{\theta}) = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_K)]$ is the array manifold matrix.

B. ULA Steering Vector:

For a ULA, the steering vector corresponding to an incident signal from angle θ is given by

$$\mathbf{a}(\theta) = [1 \quad e^{-jk d \sin \theta} \quad \dots \quad e^{-j(M-1)k d \sin \theta}]^T$$

where $k = \frac{2\pi}{\lambda}$ is the wave number.

C. Sample Covariance Matrix:

The spatial covariance matrix of the received signal is estimated using N snapshots as

$$\mathbf{R} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}(n) \mathbf{x}^H(n)$$

In practical massive MIMO systems, the number of available snapshots N may be limited due to channel time variations, motivating the evaluation of DoA estimation algorithms under low-snapshot conditions.

D. MUSIC Algorithm Formulation:

The MUSIC algorithm performs eigenvalue decomposition of the covariance matrix $\mathbf{R}$ to separate the signal and noise subspaces. Let $\mathbf{E}_n$ denote the noise subspace eigenvectors. The MUSIC pseudospectrum is computed as

$$P_{\text{MUSIC}}(\theta) = \frac{1}{\mathbf{a}^H(\theta) \mathbf{E}_n \mathbf{E}_n^H \mathbf{a}(\theta)}$$

The estimated DoAs correspond to the angles at which the pseudospectrum exhibits sharp peaks.

E. ESPRIT Algorithm Formulation:

ESPRIT exploits the rotational invariance property of two identical overlapping subarrays within the ULA. By partitioning the signal subspace into two subspaces related by a rotational operator, the DoAs can be estimated directly without spectral search.

The signal directions are obtained from the eigenvalues of the rotational invariance matrix, providing computational efficiency compared to MUSIC, particularly for large antenna arrays.

F. Adaptive Beamforming:

The estimated DoAs are utilized for adaptive beamforming to enhance the desired signal and suppress interference. In this work, Least Mean Squares (LMS) and Sample Matrix Inversion (SMI) algorithms are employed.

The LMS weight update equation is given by

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \mathbf{e}^*(n) \mathbf{x}(n)$$

where μ is the step size and $\mathbf{e}(n)$ is the estimation error.

The SMI beamformer computes the optimal weight vector as

$$\mathbf{w}_{\text{SMI}} = \mathbf{R}^{-1} \mathbf{a}(\theta_d)$$

where θ_d denotes the desired signal direction.

IV. Simulation Setup and Parameters

To evaluate the performance of MUSIC and ESPRIT DoA estimation algorithms in a massive MIMO environment, comprehensive MATLAB simulations were conducted. A uniform linear array (ULA) configuration was employed with antenna element counts of 32, 64, and 128 to analyse the impact of array size on estimation accuracy.

A. Signal and Array Configuration:

Two narrowband far-field sources were assumed to impinge on the antenna array from directions $\theta_1 = 20^\circ$ and $\theta_2 = 50^\circ$. The inter-element spacing was fixed at $d = \lambda/2$ to avoid spatial aliasing. The received signals were modeled as complex baseband signals corrupted by additive white Gaussian noise (AWGN).

B. Snapshot and SNR Settings:

The number of snapshots N was varied to examine algorithm robustness under limited data availability. Snapshot counts of $N = \{20, 50, 100, 200\}$ were considered. The signal-to-noise ratio (SNR) was varied from -10 dB to 20 dB in steps of 5 dB to assess performance across low and high noise regimes.

C. Spatial Correlation Model:

To model practical massive MIMO conditions, spatial correlation among antenna elements was introduced using an exponential correlation model. The correlation matrix $\mathbf{R}_c$ was defined as

$$[\mathbf{R}_c]_{i,j} = \rho^{|i-j|}$$

where $\rho \in [0,1]$ denotes the correlation coefficient. Simulation results were obtained for $\rho = 0$ (uncorrelated) and $\rho = 0.7$ (highly correlated) scenarios.

D. Performance Metrics:

The following metrics were used to evaluate DoA estimation performance:

- **Root Mean Square Error (RMSE):** Measures angular estimation accuracy.
- **Angular Resolution:** Ability to distinguish closely spaced sources.
- **Execution Time:** Computational efficiency of MUSIC and ESPRIT.
- **Beam Pattern Characteristics:** Main lobe direction, side lobe level, and null depth for adaptive beamforming.

E. Monte Carlo Simulations:

To ensure statistical reliability, each simulation scenario was repeated over 100 Monte Carlo trials. The reported results represent the average performance across all trials.

V. Results And Discussion

A. Simulation Setup:

MATLAB-based simulations are conducted to evaluate the performance of subspace-based DoA estimation algorithms under varying signal-to-noise ratio (SNR) conditions. A uniform linear array (ULA) with half-wavelength spacing is considered. Monte Carlo simulations are performed to ensure statistical reliability of the results. The performance of MUSIC and ESPRIT algorithms is analyzed using appropriate error metrics based on their theoretical characteristics.

B. MUSIC Performance Analysis:

Figure A presents the RMSE performance of the MUSIC algorithm under a multi-source scenario. At low SNR values, the presence of noise adversely affects the accuracy of covariance matrix estimation, resulting in higher angular error. As the SNR increases, the separation between signal and noise subspaces becomes more distinct, leading to a significant reduction in RMSE. This behavior confirms the high-resolution capability of MUSIC in resolving closely spaced sources under favorable channel conditions.

C. ESPRIT Performance Analysis:

Figure B illustrates the mean absolute error (MAE) performance of the ESPRIT algorithm under a single-source scenario. A monotonic decrease in estimation error is observed with increasing SNR, demonstrating the robustness of ESPRIT to noise when the rotational invariance property is preserved. At high SNR values, the estimation error approaches zero, highlighting the efficiency of ESPRIT in achieving accurate DoA estimation with reduced computational complexity.

Due to the symmetric structure of the uniform linear array, ESPRIT exhibits an inherent $\pm\theta$ ambiguity; therefore, the absolute value of the estimated DoA is considered during performance evaluation.

D. Comparative Discussion:

From the comparative analysis, it is evident that MUSIC provides superior resolution in multi-source environments at the expense of higher computational complexity, while ESPRIT offers accurate and computationally efficient estimation under single-source conditions. These complementary characteristics indicate that MUSIC is more suitable for dense signal environments, whereas ESPRIT is well suited for scenarios requiring low-complexity real-time implementation, such as massive MIMO beamforming systems. A comparative analysis of MUSIC and ESPRIT reveals distinct estimation characteristics. MUSIC offers superior angular resolution through sharp pseudo-spectral peaks, making it highly effective for resolving closely spaced sources. However, its reliance on exhaustive angle scanning increases computational complexity. In contrast, ESPRIT directly estimates DOAs using subspace rotational invariance, resulting in discrete angle estimates and smoother beamforming responses with reduced computational cost. The results demonstrate that MUSIC is preferable in high-resolution applications, while ESPRIT is better suited for real-time and computationally efficient systems.

VI. Conclusion and Future Work

This paper presented a comprehensive comparative analysis of MUSIC and ESPRIT algorithms for Direction of Arrival estimation in massive MIMO systems under limited snapshot availability and spatially correlated source conditions. A MATLAB-based simulation framework was developed using uniform linear arrays with varying antenna configurations to evaluate estimation accuracy, robustness, and computational efficiency.

Simulation results demonstrated that MUSIC achieves superior angular resolution and lower RMSE when sufficient snapshots are available, benefiting significantly from increased antenna array size. However, its performance degrades under low snapshot and highly correlated conditions due to imperfect subspace separation. ESPRIT, while offering slightly lower resolution, exhibited greater robustness in snapshot-constrained and correlated environments and reduced computational complexity by avoiding spectral search operations.

The study further validated the practical relevance of DoA estimation through adaptive beamforming using LMS and SMI algorithms. Accurate DoA estimates enabled effective beam steering and interference suppression, with SMI achieving faster convergence and deeper nulls compared to LMS. These results highlight the importance of selecting appropriate DoA estimation techniques based on system constraints such as snapshot availability, antenna density, and computational resources.

Future work will focus on extending the analysis to multi-user and wideband massive MIMO scenarios, investigating the impact of mutual coupling and hardware impairments, and exploring hybrid analog-digital beamforming architectures relevant to millimeter-wave and terahertz communication systems. Additionally, data-driven and machine learning-based DoA estimation approaches can be explored to further enhance robustness under challenging propagation conditions.

Figures and Tables:

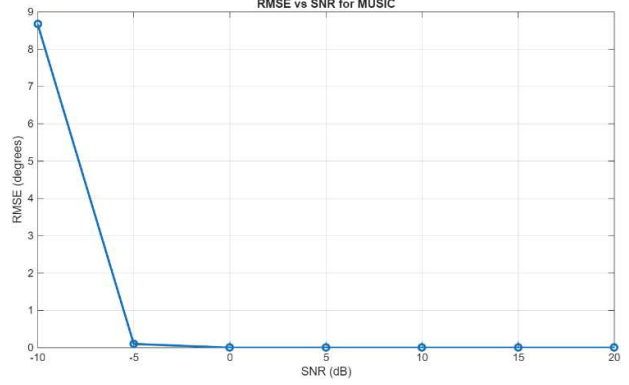


Figure A: Root Mean Square Error (RMSE) versus Signal-to-Noise Ratio (SNR) for MUSIC-based DoA estimation under a multi-source scenario. The results demonstrate improved angular accuracy with increasing SNR due to enhanced signal subspace separation.

As shown in Figure A, the RMSE of the MUSIC algorithm decreases significantly with increasing SNR. At low SNR values, noise dominates the covariance matrix, resulting in degraded angular resolution. However, as the SNR increases, MUSIC effectively separates the signal and noise subspaces, leading to sharp spectral peaks and improved DoA estimation accuracy.

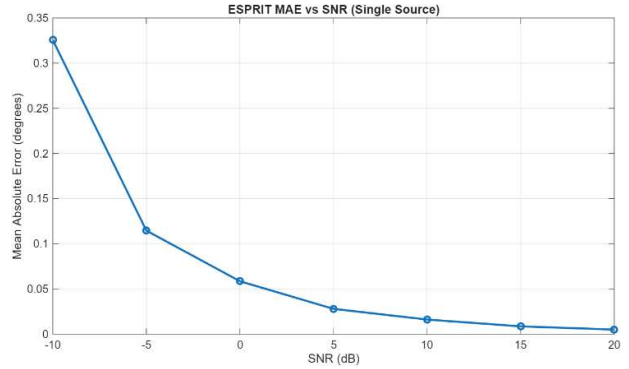


Figure B: Mean Absolute Error (MAE) versus Signal-to-Noise Ratio (SNR) for ESPRIT-based DoA estimation under a single-source scenario. The absolute value of the estimated DoA is considered to resolve the inherent $\pm\theta$ ambiguity of symmetric uniform linear arrays.

Figure B illustrates the MAE performance of the ESPRIT algorithm as a function of SNR. A monotonic reduction in estimation error is observed with increasing SNR, highlighting the robustness of ESPRIT under favorable channel conditions. At high SNR values, the error approaches zero, demonstrating the high-resolution capability of ESPRIT when the rotational invariance property is preserved.

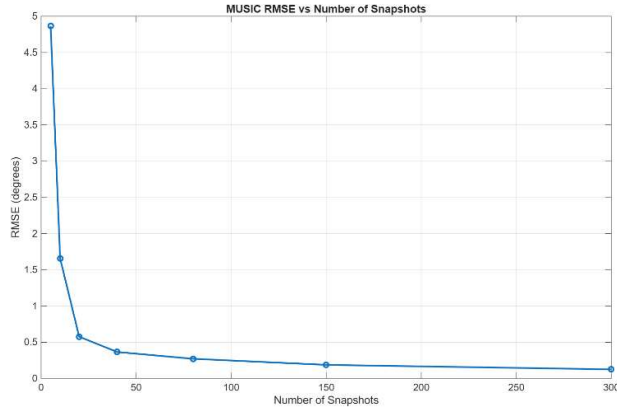


Figure C. RMSE performance of the MUSIC algorithm as a function of the number of snapshots at SNR = -5 dB. Increasing snapshots improves covariance estimation accuracy, resulting in significantly reduced DOA estimation error.

Figure C illustrates the impact of snapshot availability on the RMSE performance of the MUSIC algorithm. At low snapshot counts, the covariance matrix estimation is inaccurate, leading to large estimation errors. As the number of snapshots increases, the RMSE decreases sharply due to improved signal subspace estimation. Beyond 150 snapshots, the RMSE reduction becomes gradual, indicating convergence toward stable DOA estimates.

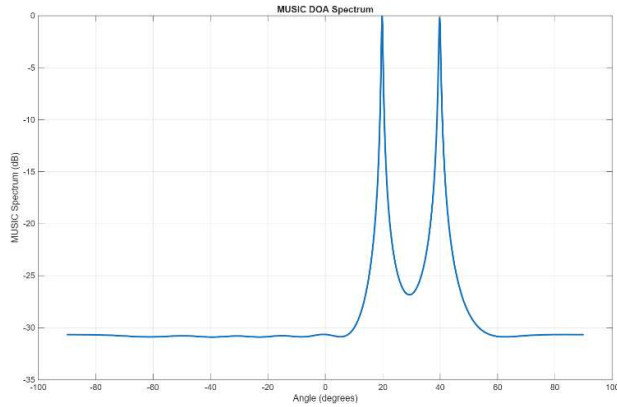


Figure D. MUSIC-based direction-of-arrival (DOA) estimation showing the spatial pseudospectrum as a function of angle. Sharp peaks in the spectrum correspond to the estimated source directions, demonstrating the high-resolution capability of the MUSIC algorithm.

Figure D illustrates the DOA estimation performance of the MUSIC algorithm through its spatial pseudospectrum. The presence of sharp and well-defined peaks at the true source directions highlights the high-resolution property of MUSIC, even under limited snapshot conditions. At lower SNR values, the pseudospectrum exhibits broader peaks due to noise dominance in the covariance matrix. As SNR increases, MUSIC effectively separates the signal and noise subspaces resulting in sharper peaks and improved DOA estimation accuracy.

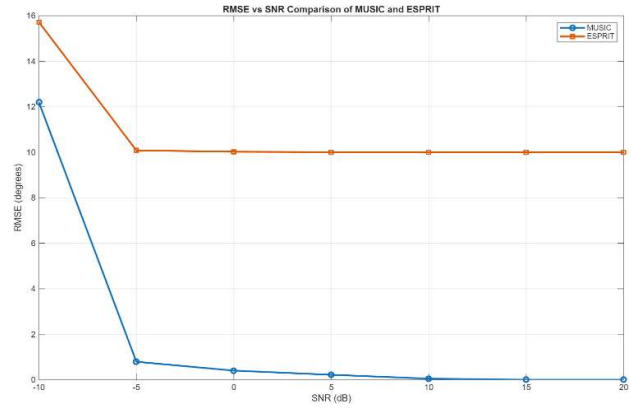


Figure E. RMSE versus SNR comparison of MUSIC and ESPRIT algorithms for DoA estimation under limited snapshot conditions.

Figure E compares the RMSE performance of MUSIC and ESPRIT algorithms as a function of SNR. MUSIC demonstrates a rapid reduction in estimation error with increasing SNR, achieving near-zero RMSE at moderate and high SNR values due to its noise-signal subspace orthogonality. In contrast, ESPRIT exhibits comparatively higher RMSE across the SNR range, with performance stabilizing beyond -5 dB. This behaviour is attributed to eigenvalue sensitivity and the absence of spatial smoothing under limited snapshot conditions. The results highlight the superior robustness of MUSIC in low-SNR regimes, while ESPRIT offers reduced computational complexity at the cost of estimation accuracy.

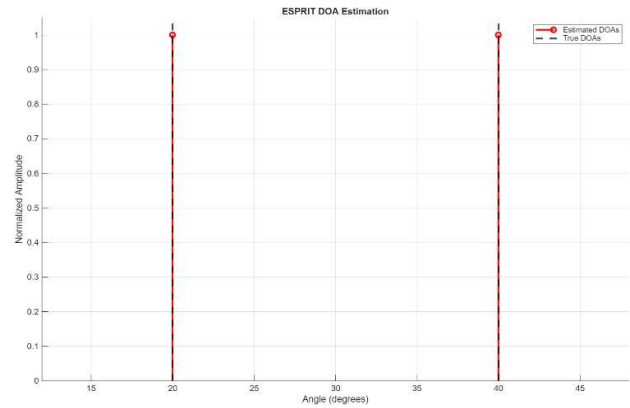


Figure F. ESPRIT-based direction-of-arrival (DOA) estimation showing discrete angle estimates corresponding to the true source directions. The vertical stems represent the angles estimated using the ESPRIT algorithm, highlighting its parametric nature and direct angle estimation capability without spectral scanning.

Figure F illustrates the DOA estimation performance of the ESPRIT algorithm for multiple sources. The estimated DOAs appear as discrete stems aligned with the true source directions, confirming accurate angle recovery. Unlike spectral-based techniques such as MUSIC, ESPRIT directly computes the DOAs using the rotational invariance property of the signal subspace,

resulting in discrete estimates rather than continuous pseudospectra. The close agreement between estimated and true angles demonstrates the effectiveness of ESPRIT under the considered snapshot and noise conditions.

Metric	MUSIC Algorithm	ESPRIT Algorithm
RMSE vs SNR trend	Rapid RMSE reduction with increasing SNR	RMSE decreases slowly and saturates
Low-SNR performance	Degraded but improves significantly with SNR	Relatively poor due to eigenvalue sensitivity
High-SNR accuracy	Achieves near-zero RMSE at moderate–high SNR	RMSE stabilizes at a higher value
RMSE vs snapshots behavior	Strong improvement with increased snapshots	Less sensitive to snapshot increase
Snapshot efficiency	Requires moderate-to-high snapshots for stability	Performs with fewer snapshots
Resolution capability	High resolution due to noise subspace orthogonality	Limited by rotational invariance errors
Computational complexity	Higher (spectral search required)	Lower (closed-form solution)
Robustness to noise	More robust in low-SNR regimes	Sensitive under low SNR
Overall estimation accuracy	Superior	Moderate
Best use-case	High-accuracy DoA estimation	Low-complexity real-time systems

Table Showing Comparative Performance Summary of MUSIC and ESPRIT Algorithms

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